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The effects of cerebellar transcranial direct current stimulation on static and dynamic postural stability in older individuals: a randomized double-blind sham-controlled study

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Keywords: balance, cerebellum, older adult, posture, transcranial direct current stimulation

Abstract

The aging population is growing rapidly. Risk of falling is higher in older people compared to young adults due to several reasons including poor posture and balance. The main aim of this study was to investigate the effect of cerebellar anodal transcranial direct current stimulation (a-tDCS) on static and dynamic postural stability in older individuals. Twenty-nine older adults participated in this study and were randomly allocated to two groups of active a-tDCS (experimental; $n = 14$) or sham tDCS group ($n = 15$). Experimental group received cerebellar a-tDCS for 20 min with intensity of 1.5 mA. Anterior–posterior and medial–lateral postural stability indices (postural sway) in addition to Berg Balance Score were measured before and after the intervention. Postural sways in static and dynamic postural tasks were significantly decreased ($P < 0.05$) after cerebellar a-tDCS, in addition to Berg Balance Score that increased significantly in active cerebellar a-tDCS group ($P < 0.05$). However, there were no significant changes in postural stability indices or Berg Balance Score in sham group ($P > 0.05$). The findings indicated immediate effect of cerebellar a-tDCS on improvement of postural control and balance in older individuals.

Introduction

As the aging population rapidly grows in the modern societies, the development of appropriate intervention strategies for enhancement of the age-related diminished performances is critical (Bernard & Seidler, 2014). It has been shown that both motor and cognitive performances deteriorate with age (Bernard & Seidler, 2014; Ehsani *et al.*, 2015). Elderly population might show poor postural control, larger postural sway and longer reaction time (Brauer *et al.*, 2002; Buatois *et al.*, 2006), which may significantly affect their gait and increase the risk of falling (Bohannon *et al.*, 1984; Maki *et al.*, 1990; Prince *et al.*, 1997; Laughton *et al.*, 2003; Holviala *et al.*, 2012).

It has been shown that cerebellar morphology changes with age (Bernard & Seidler, 2014). This observation provides insight into the potential contributions of the cerebellum to age-related motor and cognitive deteriorations (Bernard & Seidler, 2014). Some studies that assessed balance and gait deficits in older adults have indicated that older adults had significant changes in different regions of their cerebellum (Kafri *et al.*, 2013). It appears that age negatively impacts cerebellar white matter structural integrity in the cerebellar peduncles and its connecting pathways with thalamus and cortical regions (Sullivan & Pfefferbaum, 2006; Cavallari *et al.*, 2013b; Kafri *et al.*, 2013).

Cerebellum is one of the important brain areas contributing in control of motor and cognitive tasks, which is strongly interconnected with the motor and prefrontal cortices (Sullivan & Pfefferbaum, 2006; Bernard & Seidler, 2014). The cerebellar vermis considerably involves in control of the axial muscles, which play an important role in equilibrium and postural control (Shumway-Cook & Woollacott, 2012). Some studies indicated smaller cerebellar vermis volume in older adults (Raz *et al.*, 1998; Luft *et al.*, 1999; Rhyu *et al.*, 1999;

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Sullivan *et al.*, 2000). Rosano *et al.* (2007) concluded that the cause of slower walking speed or poor balance in older adults could be due to their smaller volume in cerebellum, prefrontal cortex, putamen and the posterior parietal lobe (Rosano *et al.*, 2007). In addition, aging significantly decreases cerebellar white matter structural integrity and impacts the cerebellar peduncle functions negatively (Sullivan & Pfefferbaum, 2006; Pagani *et al.*, 2008; Giorgio *et al.*, 2010). Cerebellar peduncles are the primary source of outputs from the cerebellum to the midbrain, cortex, spine, and vestibular regions (Latash, 2008). It has been suggested that age-related balance, postural control, and motor performance deficits may be correlated with changes in cerebellar white matter structural integrity and diminished cerebellar vermis volume (Rosano *et al.*, 2007; Cavallari *et al.*, 2013). Compared to cerebellum and subcortical structures, motor cortex has smaller role in postural control of healthy individuals (Prentice & Drew, 2001). Accordingly, it seems that cerebellum is a better target for application of novel interventions for enhancement of postural control in older adults (Bernard & Seidler, 2014).

Non-invasive brain stimulation (NIBS) is a promising method to modulate the cerebellum and cortico-cerebellar circuits (Luft *et al.*, 1999). Anodal transcranial direct current stimulation (a-tDCS) is a NIBS technique in clinical neuroscience that could be used to modulate brain functions (Nitsche & Paulus, 2000, 2001; Rosenkranz *et al.*, 2000; Nitsche *et al.*, 2003; Boggio *et al.*, 2006; Hummel & Cohen, 2006; Hunter *et al.*, 2009; Ehsani *et al.*, 2016c). The potential benefits of a-tDCS in rehabilitation have recently been indicated in a number of studies (Reis *et al.*, 2008; Block & Celnik, 2012; Samaei *et al.*, 2017). Moreover, behavioral benefits of cerebellar a-tDCS in both motor and cognitive tasks have also been shown (Ferrucci & Priori, 2014).

Literature indicates effectiveness of a-tDCS of primary motor cortex (M1) on enhancement of posture, balance and gait parameters in conditions such as Parkinson's disease, stroke or cerebral palsy (Geroin *et al.*, 2011; Alon *et al.*, 2012; Kaski *et al.*, 2013, 2014; Duarte *et al.*, 2014; Grecco *et al.*, 2014; Dumont *et al.*, 2015). In addition, some studies reported the positive effects of prefrontal a-tDCS on postural control and balance during dual tasking in healthy older adults (Zhou *et al.*, 2015; Manor *et al.*, 2016). Some studies assessing the polarity effect of cerebellar tDCS also indicated positive effect of cerebellar tDCS on posture and balance in healthy individuals (Inukai *et al.*, 2016; Foerster *et al.*, 2017). Due to the fact that cerebellum plays a more crucial role in maintenance of posture and balance compared to M1 (Prentice & Drew, 2001) and also by considering the significant effects of aging on the size, morphology, and function of cerebellum in older adults (Rosano *et al.*, 2007; Cavallari *et al.*, 2013; Bernard & Seidler, 2014), the cerebellum has been chosen as the stimulation site in the current study.

Although it seems that cerebellar a-tDCS may be specifically helpful for older individuals who have demonstrated deficits in their postural control (Bernard & Seidler, 2014), the effect of cerebellar a-tDCS on posture and balance of older adults has not been investigated yet. The purpose of this study was to investigate the effect of bilateral cerebellar a-tDCS on static and dynamic postural stability and balance in older adults.

We hypothesized that

1 Cerebellar a-tDCS improves static postural stability indices in older individuals.

2 Cerebellar a-tDCS improves dynamic postural stability indices in older individuals.

3 Cerebellar a-tDCS improves balance in older individuals.

4 Sham tDCS does not have any effect on static and dynamic postural stability or balance in older individuals.

Method and materials

Participants

Thirty-four healthy older adults were recruited through a convenient non-probability sampling to participate in the study. This sample size shows 85% power to declare the effect of cerebellar a-tDCS on standing balance with 95% confidence interval (CI). Figure 1 presents the flowchart of the eligibility assessment throughout the study (enrollment, allocation, follow-up, and analysis). Thirty participants met the inclusion and exclusion criteria (16 women and 14 men with mean age of 65.79 ± 6.11 years) and were randomly divided into two groups of active cerebellar and sham a-tDCS by sequences of computer-generated random numbers. Inclusion criteria were being older than 60 years old, ability to stand for 30 s without supportive devices, having a score of less than 5 in a fall risk assessment scale (Hendrich *et al.*, 2003; Ivziku *et al.*, 2011), having a score of more than 40 in a Berg Balance Scale (Berg *et al.*, 1992), not having fear of falling (Bower *et al.*, 2015), and not having any history of neurological disease or severe musculoskeletal disorders which lead to degree of disability. Exclusion criteria were having a history of any neurological disease (e.g., Parkinson's disease and Alzheimer's disease), brain surgery, brain tumor, peripheral neuropathy, visual or auditory problems, vertigo, vestibular system disorder, severe postural control disorder, having intracranial metal implantation, using medications for any neurological conditions, using psychoactive drugs and any musculoskeletal or rheumatoid condition affecting range of motion in lower extremities. All participants were interviewed and examined by a physician against inclusion and exclusion criteria before being enrolled in the study. Written informed consent was obtained from each participant. This trial study was performed based on the criteria of CONSORT checklist.

Study design

A parallel, randomized, double-blind, sham-controlled study design was utilized in current study. Twenty-nine participants participated in this study: 14 participants in the active cerebellar a-tDCS group and 15 participants in the a-tDCS sham group. The static and dynamic standing balance of the participants was evaluated before and immediately after receiving 20 min of cerebellar a-tDCS by a Biodex Balance System (BBS). The balance was assessed before and 48 h after receiving the tDCS intervention by Berg Balance Scale. Two researchers were involved in this study. One of them administered the a-tDCS applications, and the second one, blinded to the treatment groups, was responsible for assessment of outcome measures.

The study was approved by Human Ethics Committee at the Semnan University of Medical Sciences, Semnan, Iran, which is associated with Declaration of Helsinki. The study was registered as a clinical trial study on Iranian Registry of Clinical Trials (The registration number: IRCT2016010121294N3, www.irct.ir).

Transcranial direct current stimulation

A tDCS device (ActivaDose II, Manufactured by ActivaTeK™ Inc., Taiwan) with active electrode and returning electrode size of 5 × 5 cm covered by saline-soaked surface sponges were used to deliver 1.5 mA a-tDCS for 20 min over cerebellum (Hardwick *et al.,* 2014; Ferrucci *et al.*, 2015; Priori *et al.*, 2014). To minimize the side effects, the first and the last 10 s of stimulation were gradually adjusted (Nitsche & Paulus, 2000; Galea *et al.*, 2009; Brunoni *et al.*, 2011). In the experimental group, the active (anode) and the returning (cathode) electrodes were placed over the cerebellum

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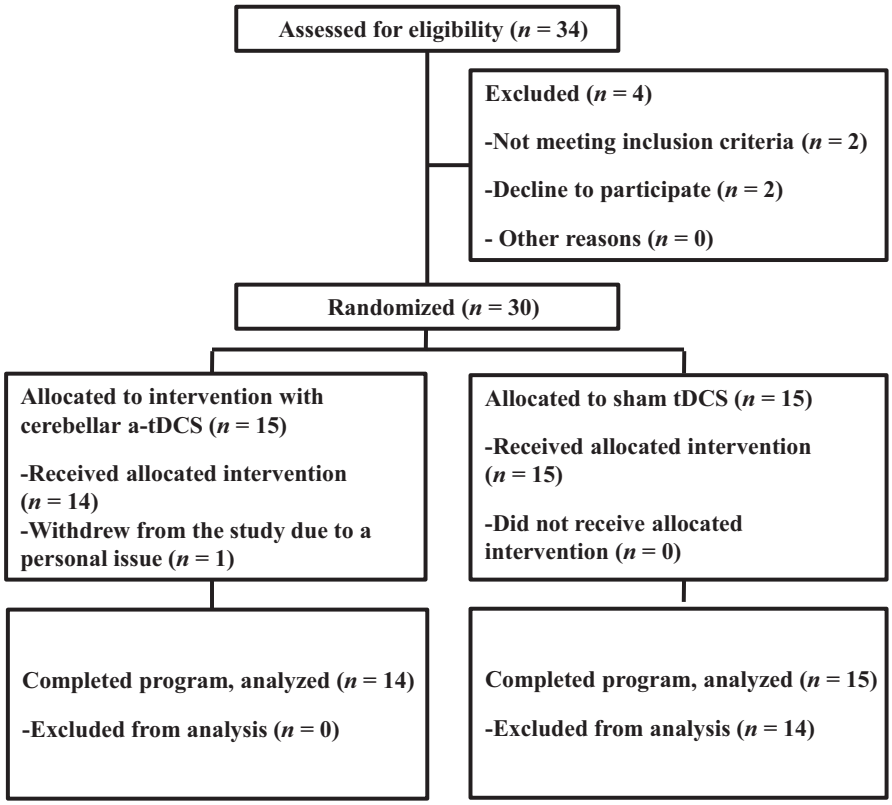

FIG. 1. Flow diagram of participant's eligibility assessment.

(1 cm below inion of occipital bone) and right arm, respectively (Ferrucci *et al.*, 2008, 2013; Ehsani *et al.*, 2016c; Samaei *et al.*, 2017; Fig. 2). A 5 × 5 cm active electrode enabled us to directly stimulate vermis, intermediate hemispheres and to some extent cerebellar hemispheres.

The fade-in—short stimulation—fade-out (FiSsFo) approach was used in sham tDCS group to induce the assumed initial cutaneous sensations. Similarly, the electrodes were placed in the same positions as the active tDCS group, and the 1.5 mA tDCS was slowly turned off after 30 s stimulation (Ferrucci *et al.*, 2008, 2013). The FiSsFo approach is a reliable method to keep the blinding integrity (Gandiga *et al.*, 2006).

Assessment of balance and postural stability indices

Berg Balance Scale (with a maximum score of 56) is a valid and reliable scale, which was used to assess balance during static and dynamic activities of varying difficulties (Berg *et al.*, 1992). The BBS (#950-302, SD, New York) is a valid and reliable stabilometric device for objective assessment of balance and postural stability on stable or unstable surfaces (Schmitz & Arnold, 1998). The BBS was used to measure multi-directional postural sway of the participant during standing on both static and dynamic platforms with varying degrees of instability (12 levels). The platform stability of BBS has a static level and 12 dynamic (movable) levels (1–12), from the greatest to the lowest instabilities, respectively (Schmitz & Arnold, 1998). The results of a pilot testing of the BBS indicated that the participants were not able to complete dynamic tests of stability level 4 or lower for 30 s, without losing their balance. Therefore, the static level and dynamic level of 5, as the lowest instability level in this study, were used to test static and dynamic conditions in the study, respectively. Participants were instructed to stand on the platform of BBS level, without touching the handrails during the test, while they were barefoot and their arms crossed on their chest (Fig. 3). All assessments were conducted with eyes 'open' for a period of 30 s (Arnold & Schmitz, 1998; Sherafat *et al.*, 2013; Ehsani *et al.*, 2016a,b). We carried out two balance assessments, randomly: (i) standing on the stable platform, and (ii) standing on the movable platform (level 5). The anterior–posterior stability index (APSI), medial–lateral stability index (MLSI), and overall stability index (OSI) could be calculated using the following formulas, respectively:

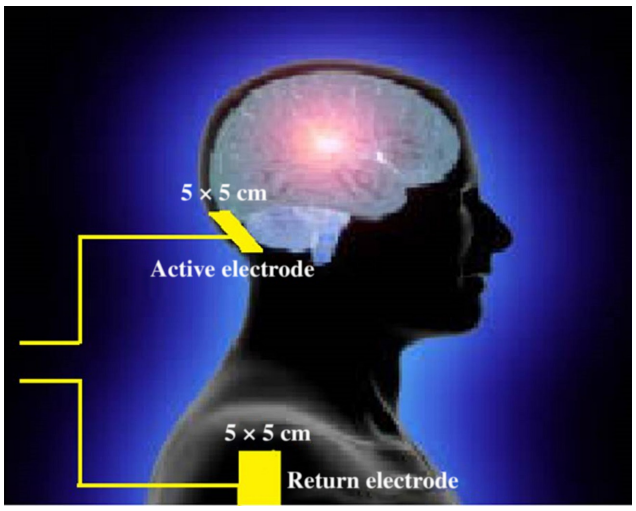

FIG. 2. Electrode montages for a-tDCS over cerebellum; active electrode was placed over the cerebellum (1 cm below inion of occipital bone), and the returning electrode was placed over right arm. [Colour figure can be viewed at wileyonlinelibrary.com].

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FIG. 3. Standing on the platform level of Biodex Balance System during postural test. [Colour figure can be viewed at wileyonlinelibrary.com].

$$\text{APSI} = \left(\sqrt{\frac{\sum (0-Y)^2}{\text{number of samples}}}\right),$$

that is, Y = platform displacement in the sagittal plan.

$$\text{MLSI} = \left(\sqrt{\frac{\sum (0-X)^2}{\text{number of samples}}}\right),$$

that is, X = platform displacement in the frontal plan.

$$\text{OSI} = \left(\sqrt{\frac{\sum (0-X)^2 + (0-Y)^2}{\text{number of samples}}}\right),$$

that is, X = platform displacement in the sagittal plan, Y = platform displacement in the sagittal plan.

Data of postural stability in different directions were obtained by BBS after completing each test. APSI, MLSI, and OSI represent platform displacements in the sagittal, frontal, and a combination of both planes. A high score in these indices may indicate poor postural control (Schmitz & Arnold, 1998). On the other hand, reduction in these indices following the intervention may indicate improvement in postural control (Schmitz & Arnold, 1998).

Experimental procedures

Participants were asked to randomly stand on the static and dynamic platforms of BBS and maintain balance for 30 s (Sherafat *et al.*, 2013; Ehsani *et al.*, 2016a,b). After performing balance tests and saving the BBS baseline data, participants were asked to sit on a chair to receive a-tDCS intervention for 20 min (Grecco *et al.*, 2014). Then, the balance assessments by BBS were randomly repeated immediately after the intervention to obtain BBS intervention data. In addition, the Berg Balance Scale was completed after 48 h after intervention. Participants were blinded to the sham or active conditions of a-tDCS.

Assessment of the side effects

A questionnaire regarding the side or adverse effects of cerebellar a-tDCS was completed during and after cerebellar a-tDCS intervention. The questionnaire included rating and numeric analogue scales (NAS; e.g., 0 = no tingling to 10 = worst tingling imaginable). The existence and the severity of side effects such as itching, tingling, or burning sensations under the electrodes were determined by this scale (Nitsche *et al.*, 2008; George & Aston-Jones, 2010).

Data analysis

SPSS software, version 22, was used to analyze the data. Shapiro–Wilk test was used to assess the normality of distribution for tested variables. All variables of the study had normal distribution. An independent sample *t*-test was used to assess the baseline values between the groups. A three-way repeated-measures ANOVA was used to assess the main effects of group (cerebellar and sham a-tDCS), time (before and after receiving a-tDCS), postural stability (static or dynamic levels of BBS) or balance (Berg Balance Score), and their interactions on the postural stability indices. *Post hoc* tests with Bonferroni corrections were carried out as required. Type I error (α) was set at 0.05, and the power of tests was considered 0.85.

Results

The effect of a-tDCS of cerebellum on postural stability indices and balance

The demographic details and the baseline data for the participants in each group are presented in Table 1. There were no significant differences in age, gender, body mass index, or baseline APSI, MLSI, OSI, or Berg Balance Scores between groups ($P > 0.05$). The results of the three-way repeated-measures ANOVA are shown in Table 2. The between-subjects main effect of groups is significant ($P < 0.001$). The within-subjects main effect of postural stability indices and dynamicity is also significant ($P < 0.001$), which indicates significant differences in postural stability indices in different dynamicity of postural task and time (before and after) in each group. ANOVA also shows that there are significant interaction effect between groups and postural stability indices ($P < 0.05$), which

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TABLE 1. Demographic data and baseline values for the participants in experimental and sham groups (mean ± SEM)

Variable	Group (Mean ± SEM)		P value
	Cerebellar a-tDCS	Sham tDCS	
Age (year)	66.08 ± 6.33	65.50 ± 6.14	0.82
Gender (Male/Female)	6/8	7/8	0.62
BMI	24.92 ± 0.98	24.17 ± 1.12	0.44
APSI (S)	1.16 ± 0.19	1.05 ± 0.09	0.65
MLSI (S)	1.00 ± 0.11	0.93 ± 0.09	0.64
OSI (S)	1.17 ± 0.19	0.94 ± 0.10	0.29
APSI (D)	2.58 ± 0.18	2.42 ± 0.19	0.12
MLSI (D)	2.38 ± 0.13	2.03 ± 0.21	0.44
OSI (D)	3.39 ± 0.25	3.11 ± 0.26	0.82
Berg Balance Score	42.45 ± 1.43	42.91 ± 1.87	0.56

SEM, Standard error of measurement; BMI, body mass index; APSI, anterior –posterior stability index; S, static; MLSI, medial–lateral stability index; OSI, overall stability index; D, dynamic.

indicates significant differences in postural stability indices between groups based on intervention. *Post hoc* analysis using Bonferroni correction showed significant APSI, MLSI, and Berg Balance Score changes after active cerebellar a-tDCS (Table 3). Whereas in sham group, there were no significant APSI, MLSI, OSI, and Berg Balance Score changes after receiving sham tDCS (Table 3). In addition, independent *t*-test has shown significant Berg Balance Score and postural stability index changes after intervention between groups (Figs 4 and 5). The anterior–posterior and medial–lateral postural sway in static and dynamic postural tasks were significantly decreased after cerebellar a-tDCS (Fig. 4). In addition, Berg Balance Score was significantly increased after intervention in cerebellar a-tDCS group (Fig. 5).

Safety and side effects of a-tDCS

All participants tolerated the a-tDCS very well in both groups. They all reported minimal adverse or side effects during or after the

TABLE 2. ANOVA results for the effects of a-tDCS on postural stability indices

Variables	Effect		Df	F	P value
APSI	Main	Group	1	4.45	0.03*
		Evaluation	1	11.53	0.003*
		Dynamicity	1	142.78	< 0.001*
	Interaction	Group × Evaluation	1	5.12	0.034*
		Group × Dynamicity	1	0.48	0.49
		Evaluation × Dynamicity	1	2.06	0.16
		Group × Dynamicity × Evaluation	1	0.57	0.46
MLSI	Main	Group	1	6.55	0.02*
		Evaluation	1	6.12	0.02*
		Dynamicity	1	84.34	< 0.001*
	Interaction	Group × Evaluation	1	27.01	< 0.001*
		Group × Dynamicity	1	0.18	0.68
		Evaluation × Dynamicity	1	0.25	0.62
		Group × Dynamicity × Evaluation	1	4.53	0.05
OSI	Main	Group	1	4.02	0.04*
		Evaluation	1	5.73	0.03*
		Dynamicity	1	165.33	< 0.001*
	Interaction	Group × Evaluation	1	1.12	0.30
		Group × Dynamicity	1	0.71	0.41
		Evaluation × Dynamicity	1	5.30	0.03*
		Group × Dynamicity × Evaluation	1	1.37	0.26

APSI, Anterior–posterior stability index; S, static; MLSI, medial–lateral stability index; OSI, overall stability index,* indicates significant difference.

TABLE 3. *Post hoc* pair-wise comparison of postural stability indices before and after intervention in each group

Group	Stability levels	Variables	Mean differences (95% CI)	P value	Effect Size
Cerebellar a-tDCS	Static	APSI	0.60 (0.13 to 1.07)	0.018*	0.90
		MLSI	0.51 (0.19 to 0.82)	0.004*	0.95
		OSI	0.19 (−0.5 to 0.85)	0.57	0.38
	Dynamic	APSI	1.11 (0.23 to 1.99)	0.018*	0.93
		MLSI	0.99 (0.42 to 1.59)	0.003*	0.91
		OSI	1.08 (0.09 to 2.08)	0.036	0.88
	Berg Balance Score		−3.97 (−5.26 to −3.19)	0.013*	2.34
Sham-control	Static	APSI	0.09 (−0.19 to 0.38)	0.19	0.36
		MLSI	−0.12 (−0.45 to 0.22)	0.49	0.41
		OSI	0.10 (−0.05 to 0.26)	0.46	0.46
	Dynamic	APSI	0.25 (−0.32 to 0.82)	0.36	0.41
		MLSI	−0.29 (−0.52 to −0.16)	0.08	0.44
		OSI	0.39 (−0.27 to 1.06)	0.22	0.47
	Berg Balance Score		−0.44 (−0.61 to −0.11)	0.11	0.25

CI, Confidence interval; APSI, anterior–posterior stability index; S, static; MLSI, medial–lateral stability index; OSI, overall stability index, * indicates significant difference.

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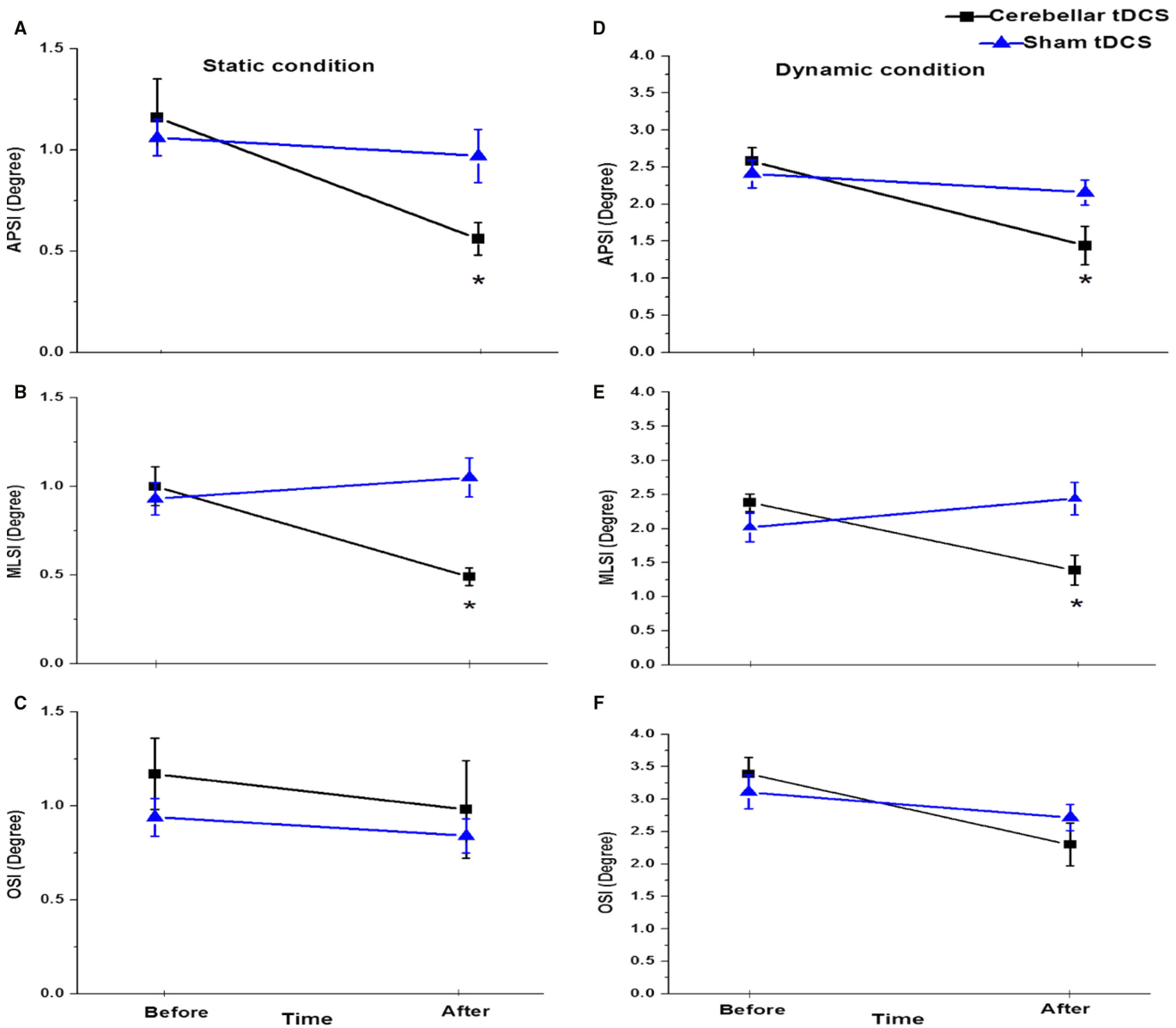

FIG. 4. The comparison of postural indices (mean ± SEM) before and after intervention between groups in static and dynamic testing conditions, (A) anterior–posterior stability index (APSI; mean ± SEM) in static condition, (B) medial–lateral stability index (MLSI; mean ± SEM) in static condition, (C) overall stability index (OSI; mean ± SEM) in static condition, (D) APSI (mean ± SEM) in dynamic condition, (E) MLSI (mean ± SEM) in dynamic condition, (F) OSI (mean ± SEM) in dynamic condition; * indicates significance difference, SEM, standard error of measurement. [Colour figure can be viewed at wileyonlinelibrary.com].

intervention (Table 4). Itching was the most reported side effect of a-tDCS. Participants did not report any side effect after the end of stimulation. In addition, participants did not report any burning sensation, headache, or pain during or after stimulation.

Discussion

The current study for the first time evaluated the effect of one-session cerebellar a-tDCS on postural stability of older adults and indicated that cerebellar a-tDCS could improve postural stability indices and balance in older individuals. The technique was well tolerated by all participants. The findings in the current study indicated significant improvement in postural stability and balance after active cerebellar a-tDCS, but not sham stimulation.

We hypothesized that static postural stability indices would be improved by cerebellar a-tDCS in the older adults. The findings of the current study supported this hypothesis and indicated that participants who received cerebellar a-tDCS had significantly less postural sway in AP and ML directions in the static conditions, which means a better static postural control. Due to lack of comparable studies in the literature, the findings in the current study will only be compared with a-tDCS studies which investigated static postural stability of the individuals with postural disorders (Duarte *et al.*, 2014; Grecco *et al.*, 2014; Dumont *et al.*, 2015). In line with the findings in the current study, literature supports positive effect of a-tDCS of M1 on the static postural control (Duarte *et al.*, 2014; Grecco *et al.*, 2014; Dumont *et al.*, 2015). Duarte *et al.* (2014) have shown that gait training on a treadmill combined with M1 a-tDCS improved static balance which was demonstrated by reduction in AP and ML sway and improvement in functional performance in children with cerebral palsy (Duarte *et al.*, 2014). The effects in this study lasted for a month after the completion of the intervention (Duarte *et al.*,

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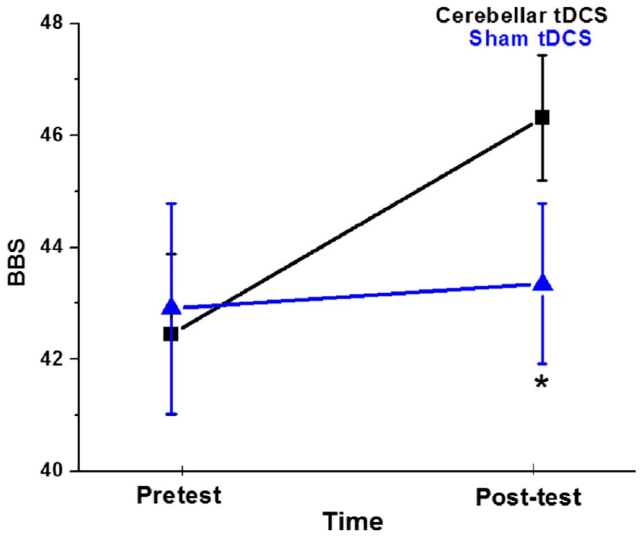

FIG. 5. The comparison of Berg Balance Score (mean ± SEM) before and after intervention between groups; * indicates significance difference, SEM, error of measurement. [Colour figure can be viewed at wileyonlinelibrary.com].

2014). According to the evidence, the cerebellum has rich connections with the other important regions of brain that leads to have a significant contribution in controlling cognitive and motor tasks, balance, and postural control in human (Sullivan & Pfefferbaum, 2006; Bernard & Seidler, 2014). As the function and structure of the cerebellum can be changed during the aging process (Bernard & Seidler, 2014), potentially the static or dynamic postural control and balance can be deteriorated in the older adults which in turn can increase the risk of falling in this population (Maki *et al.*, 1990; Prince *et al.*, 1997; Laughton *et al.*, 2003).

We also hypothesized that dynamic postural stability indices and balance would be improved by cerebellar a-tDCS in the older individuals. The findings in the current study supported this hypothesis and indicated that cerebellar a-tDCS could significantly decrease postural sway in AP and ML directions in the dynamic condition of tests ($P < 0.05$) and increase Berg Balance Score ($P < 0.05$). Similar to our findings, literature supports a positive effect of prefrontal tDCS on dynamic balance and walking performance during dual tasks in older adults (Zhou *et al.*, 2015; Manor *et al.*, 2016). Manor *et al.* (2016) showed that application of one-session prefrontal tDCS could decrease the cost of dual tasking in older adults and improve their walking and postural control (Manor *et al.*, 2016). Zhou *et al.* (2015) also investigated the effect of one-session prefrontal tDCS on center of pressure (COP) fluctuations during single and dual tasks and showed increased COP complexity and postural control improvement during tDCS intervention in older adults (Zhou *et al.*, 2015). A number of some studies also investigated the effect of cerebellar tDCS on posture and balance in healthy individuals (Inukai *et al.*, 2016; Foerster *et al.*, 2017). Similar to our study, static and dynamic postural stability were assessed by BBS before and immediately after cerebellar tDCS in healthy individual (Foerster *et al.*, 2017). This study reported static postural control changes and increased overall PSI by cerebellar cathodal tDCS in healthy individuals (Foerster *et al.*, 2017). It seems that cerebellar tDCS can modulate cerebellar–motor cortex connectivity and positively affect motor, cognitive and postural control processing (Celnik, 2015). There is evidence that vermis volume and cerebellar white matter structural integrity are impacted by aging (Raz *et al.*, 1998; Luft *et al.*, 1999; Rhyu *et al.*, 1999; Sullivan *et al.*, 2000; Sullivan & Pfefferbaum, 2006; Pagani *et al.*, 2008; Kafri *et al.*, 2013). Cerebellar white matter tracts are the primary connectors for the cerebellum to the other brain regions, and vermis is one of the key regions for balance and postural control (Latash, 2008; Shumway-Cook & Woollacott, 2012). Cerebellar a-tDCS may improve the function of vermis region or white matter tracts by increasing Purkinje cell activation (Celnik, 2015). Purkinje cell activation inhibits the deep cerebellar nucleus neurons to produce the appropriate motor output and suppress unwanted activity (Celnik, 2015). Indeed, it appears that cerebellar a-tDCS facilitates the connectivity of cerebellum with the other brain regions and increases the cerebellar controlling functions on motor cortex, vestibular system, midbrain, and the other brain regions (Priori *et al.*, 2014; Rahmati *et al.*, 2014). Accordingly, this intervention may improve the cerebellar functions, modulate cognitive, motor, and balance performance, and compensate for age-related structural changes in cerebellar region in older adults.

TABLE 4. Evaluation of side effects during tDCS intervention in both groups (numeric sensation scores)

	Anode electrode		Cathode electrode	
	Cerebellar a-tDCS	Sham tDCS	Cerebellar a-tDCS	Sham tDCS
Tingling sensation				
Beginning	4.92 ± 0.75	2.11 ± 0.51	0.87 ± 0.34	0.66 ± 0.37
Middle	3.85 ± 0.41	1.70 ± 0.33	0.75 ± 0.39	0.60 ± 0.19
End	2.01 ± 0.37	0.75 ± 0.09	0.52 ± 0.18	0.43 ± 0.12
Itching sensation				
Beginning	3.45 ± 0.41	1.19 ± 0.17	0.92 ± 0.25	0.85 ± 0.17
Middle	2.85 ± 0.29	0.85 ± 0.21	0.73 ± 0.33	0.38 ± 0.15
End	1.78 ± 0.32	0.67 ± 0.13	0.70 ± 0.21	0.19 ± 0.11
Burning sensation				
Beginning	–	–	–	–
Middle	–	–	–	–
End	–	–	–	–
Pain				
Middle	–	–	–	–
End	–	–	–	–
Headache				
Beginning	–	–	–	–
Middle	–	–	–	–
End	–	–	–	–
Not tolerated				
Beginning	–	–	–	–
Middle	–	–	–	–
End	–	–	–	–
Others				
Beginning	–	–	–	–
Middle	–	–	–	–
End	–	–	–	–

Numeric Analog Scale (NAS): 0 is rated as no sensation and 10 rated as the worst sensation imaginable. The participants reported their sensations under both active (anode) and return (cathode) electrodes during two phases of tDCS stimulation of cerebellum and sham tDCS: beginning (0–7 min of stimulation), middle (7–14 min of stimulation), end (14–20 min of stimulation). Scores are reported as mean ± SEM (SEM, standard error of measurement).

The findings of the current study for the first time confirmed the positive effects of one-session cerebellar a-tDCS on static and dynamic postural stability in healthy older adults. In addition, the increases in Berg Balance Scores (CI: 3.19–5.26) in cerebellar a-tDCS group were significantly more than sham tDCS group during 48-h follow-up (CI: 0.11–0.61), which indicated important clinical findings for balance controlling in older adults.

Finally, it was hypothesized that static and dynamic postural stability indices would not change following application of sham a-

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tDCS. Our findings supported this hypothesis and showed no significant changes in static and dynamic postural stability in the sham group. In line with the findings in current study, a number of studies, which investigating the effect of sham tDCS on posture, balance, and gait of the individuals with postural disorder, did not observe any effect on the balance, postural control, and gait performance (Geroin *et al.*, 2011; Sohn *et al.*, 2013; Grecco *et al.*, 2014; Tahtis *et al.*, 2014). This finding indicates that cerebellar a-tDCS is responsible for the observed changes and placebo effect did not have any role on this regard.

Safety and side effects of a-tDCS

The findings in current study demonstrated that cerebellar a-tDCS technique is a safe method with minimal side effects and no adverse effects in older individuals. General discomfort (itching and/or tingling) was reported as the common side effects of active and sham tDCS by older participants.

Limitations and suggestions for future research

The participants in the current study were older adults; therefore, the results cannot be generalized to young or middle-aged adults. Further studies are warranted to investigate the effect of cerebellar a-tDCS on the postural stability in young and middle-aged adults. This study can also be repeated on older individuals with high fall risk. The current study evaluated only cerebellar a-tDCS on static and dynamic postural stability. It is recommended conducting further studies to investigate the priming effects of cerebellar a-tDCS on postural training and other training techniques on postural stability in older adults. In addition, the current study investigated only the immediate effects of single-session cerebellar a-tDCS on postural stability. Multiple sessions and long-term effect of cerebellar a-tDCS on postural stability of older adults can also be investigated. Moreover, center of pressure motion parameters was not assessed for assessment of postural control in the current study. Therefore, further research using force plate for assessment of center of pressure is recommended.

Conclusion

The findings in the current study for the first time confirm that one session of cerebellar a-tDCS significantly improves postural stability and balance in older individuals. The findings also indicate that these effects are not due to the placebo effects of cerebellar a-tDCS.

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Conflict of interest

There was no conflict of interest.

Author contributions

All authors Fatemeh Ehsani, Afshin Samaei, Maryam Zoghi, Rozita Hedayati and Shapour Jaberzadeh substantially contributed to the conception or design of the work or the acquisition, analysis, or interpretation of data for the work; drafting the work or revising it critically for important intellectual content; final approval of the version to be published; and agreement to be accountable for all aspects of the work.

Data accessibility

Data will become accessible by corresponding author for whoever requires.

Abbreviations

APSI, anterior–posterior stability index; a-tDCS, anodal transcranial direct current stimulation; BBS, Biodex Balance System; BMI, body mass index; CI, confidence interval; M1, primary motor cortex; MLSI, medial–lateral stability index; NAS, numeric analogue scales; NIBS, non-invasive brain stimulation; OSI, overall stability index.

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